

Muhammad Irfan

Computer Vision & Machine Learning Engineer

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No visa sponsorship required

SUMMARY

Computer vision and machine learning engineer with 10+ years of research experience and current automotive industry experience developing, optimizing, and deploying deep learning systems across manufacturing and medical imaging. At General Motors, builds image and signal models and transitions proof-of-concept solutions into optimized real-time inference workloads using NVIDIA TensorRT and Jetson Thor, backed by patents and peer-reviewed publications.

EXPERIENCE

Machine Learning Engineer General Motors R&D

Jan 2026 – Present - Warren, MI

- Develop and deliver real-time computer vision and deep learning solutions from requirements through production deployment.
- Build CNN-, RNN-, and GAN-based models for complex image and signal data, including feature extraction, classification, detection, and prediction workflows.
- Optimize deep learning models with NVIDIA TensorRT and deploy inference workloads on NVIDIA Jetson Thor, balancing accuracy, latency, throughput, memory, and computational cost.
- Design reproducible model and inference pipelines that move proof-of-concept systems toward scalable operational deployment; current programs are in the deployment phase.
- Mentor 4–6 engineers, review technical designs and deliverables, resolve implementation risks, and collaborate with cross-functional engineering teams in the United States and India.
- Lead inventor on a GM U.S. provisional patent application and a separate confidential AI/ML innovation retained as a GM trade secret.

Research Assistant

SMILES Lab, University of Michigan-Flint

Jan 2022 – Dec 2025 - Flint, MI

- Built reproducible 3D computer vision pipelines for MRI, MRA, DSA, and clinical data, covering preprocessing, training, fine-tuning, validation, error analysis, and documentation.
- Developed detection, segmentation, and risk-prediction models using CNNs, UNet variants, SegFormer, transformers, large vision models, vision-language models, and image-text-tabular fusion.
- Applied one-shot, few-shot, and federated learning to improve generalization, data efficiency, and privacy across limited and distributed datasets.
- Produced patented and peer-reviewed computer vision research through collaborations spanning the United States, Norway, the UAE, Pakistan, and South Korea.

Research Assistant

Mixed Reality & Interaction Lab, Sejong University

Sep 2018 – May 2021 - Seoul, South Korea

- Designed and evaluated deep learning systems for image enhancement, dehazing, object detection, segmentation, saliency detection, motion

SKILLS

COMPUTER VISION

Object Detection Image Classification
Segmentation Feature Extraction
Image Enhancement Motion Analysis
3D Imaging Video Analytics
Vision-Language Models

EDGE AI & DEPLOYMENT

Jetson Thor TensorRT
Real-Time Inference Embedded Systems
Model Optimization PoC-to-Production

PROGRAMMING & FRAMEWORKS

Python ROS2 MATLAB PyTorch
TensorFlow Keras Scikit-learn
OpenCV MONAI NumPy Pandas

MODELS & METHODS

LLMs Transformers RNNs CNNs
UNet variants SegFormer
Few-Shot Learning Federated Learning

PLATFORMS & TOOLS

AWS Azure GCP Git DICOM ITK
VTK 3D Slicer

EDUCATION

PhD in Computer Science

University of Michigan-Flint

Expected Dec 2026 - Flint, MI

Focus: Brain aneurysm detection, segmentation, and rupture prediction using DSA, MRA, and CT imaging.

MS in Computer Engineering

Sejong University

Aug 2020 - Seoul, South Korea

Thesis: FoV selection in 360-degree video.

BS in Computer Science

Islamia College University

Aug 2018 - Peshawar, Pakistan

Thesis: RPI-based CNN self-driving car.

estimation, image-quality assessment, and sensor-based activity recognition.

- Developed CNN-based perception and tracking methods for 360-degree video and immersive industrial environments, including field-of-view selection and abnormal-activity detection.
- Built experimentation and data pipelines, optimized models for accuracy and computational efficiency, and contributed to granted patents and peer-reviewed publications.

Research Assistant

Digital Image Processing Lab, Islamia College University

Jun 2016 – Jul 2018 - Peshawar, Pakistan

- Developed a real-time CNN-assisted autonomous-driving prototype on Raspberry Pi, integrating computer vision models with resource-constrained embedded hardware.
- Implemented and optimized lightweight motion-detection and image-enhancement models for real-time edge applications.

PATENTS & INTELLECTUAL PROPERTY

- General Motors confidential AI/ML invention, U.S. provisional patent application filed in 2026; lead inventor.
- General Motors confidential AI/ML innovation retained as a protected trade secret in 2026; lead inventor.
- U.S. Provisional Application No. 30275/70862P, “Adaptive Learning for Multi-Modal Intracranial Aneurysm Rupture Criticality Analysis,” filed 2025; lead inventor.
- KR Patent 10-2284913 B1, “User’s Field of View Selection Method and Apparatus in 360-Degree Videos,” granted 2021; lead inventor.
- KR Patent 10-2168438 B1, “Method and Server for Detecting Abnormal Activity Using Drones,” granted 2020; lead inventor.

SELECTED PUBLICATIONS

- Irfan, M. et al. (2025). One-Shot Learning for Generalization in Medical Image Classification Across Modalities. Computerized Medical Imaging and Graphics.
- Irfan, M. et al. (2024). Federated Fusion Learning with Attention Mechanism for Multi-Client Medical Image Analysis. Information Fusion.
- Irfan, M. et al. (2023). StrokeNet: Automated Segmentation and Rupture Risk Prediction of Intracranial Aneurysms. Computerized Medical Imaging and Graphics.
- Irfan, M. et al. (2021). Deep Learning-Based User Field of View Selection in 360-Degree Videos for Industrial Environments. IEEE Internet of Things Journal.
- Irfan, M. et al. (2021). Light-DehazeNet: Lightweight CNN for Single Image Dehazing. IEEE Transactions on Image Processing.
- Sajjad, M., Irfan, M. et al. (2020). Efficient and Scalable Simulation Model for Autonomous Vehicles with Economical Hardware. IEEE Transactions on Intelligent Transportation Systems.